

Mathematics

Stage 9

Paper 2

2022

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidance on marking**Difference in printing**

It is suggested that schools check their printed copies for differences in printing that may affect the answers to the questions, for example in measurement questions.

Brackets in mark scheme

When brackets appear in the mark scheme this indicates extra information that is not required but may be given.

For example:

Question	Answer	Mark	Part marks	Guidance
5	19.7 or 19.6(58...)	1		

This means that 19.6 is an acceptable truncated answer even though it is not the correct rounded answer.

The ... means you can ignore any numbers that follow this; you do not need to check them.

Accept

- any correct rounding of the numbers in the brackets, e.g. 19.66
- truncations beyond the brackets, e.g. 19.65

Do not accept

- 19.68 (since the numbers in brackets do not have to be present but if they are they should be correct).

These tables give general guidelines on marking learner responses that are not specifically mentioned in the mark scheme. Any guidance specifically given in the mark scheme supersedes this guidance.

Number and place value

The table shows various general rules in terms of acceptable decimal answers.

Accept
Accept omission of leading zero if answer is clearly shown, e.g. .675
Accept trailing zeros, unless the question has asked for a specific number of decimal places or significant figures, e.g. 0.7000
Accept a comma as a decimal point if that is the convention that you have taught the learners, e.g. 0,638

Units

For questions involving quantities, e.g. length, mass, money, duration or time, correct units must be given in the answer. Units are provided on the answer line unless finding the units is part of what is being assessed.

The table shows acceptable and unacceptable versions of the answer 1.85 m.

	Accept	Do not accept
If the unit is given on the answer line, e.g. m	Correct conversions, provided the unit is stated unambiguously, e.g.185 cm..... m (this is unambiguous since the unit cm comes straight after the answer, voiding the m which is now not next to the answer)	185..... m1850..... m etc.
If the question states the unit that the answer should be given in, e.g. 'Give your answer in metres'	1.85 1 m 85 cm	185; 1850 Any conversions to other units, e.g. 185 cm

Money

In addition to the rules for units, the table below gives guidance for answers involving money. The table shows acceptable and unacceptable versions of the answer \$0.30

	Accept	Do not accept
If the amount is in dollars and cents, the answer should be given to two decimal places	\$0.30 For an integer number of dollars it is acceptable not to give any decimal places, e.g. \$9 or \$9.00	\$0.3 \$09 or \$09.00
If units are not given on the answer line	Any unambiguous indication of the correct amount, e.g. 30 cents; 30 c \$0.30; \$0-30; \$0=30; \$00:30	30 or 0.30 without a unit \$30; 0.30 cents Ambiguous answers, e.g. \$30 cents; \$0.30 c; \$0.30 cents (as you do not know which unit applies because there are units either side of the number)
If \$ is shown on the answer line	All unambiguous indications, e.g. \$.....0.30.....; \$.....0-30.....; \$.....0=30.....; \$.....00:30.....	\$.....30..... Ambiguous answers, e.g. \$.....30 cents.....; \$.....0.30 cents..... unless units on the answer line have been deleted, e.g. \$.....30 cents.....
If cents is shown on the answer line	30.....cents	0.30.....cents Ambiguous answers, e.g.\$30cents;\$0.30cents unless units on the answer line have been deleted, e.g.\$0.30.....cents

Duration

In addition to the rules for units, the table below gives guidance for answers involving time durations. The table shows acceptable and unacceptable versions of the answer 2 hours and 30 minutes.

Accept	Do not accept
Any unambiguous indication using any reasonable abbreviations of hours (h, hr, hrs), minutes (m, min, mins) and seconds (s, sec, secs), e.g. 2 hours 30 minutes; 2 h 30 m; 02 h 30 m	Incorrect or ambiguous formats, e.g. 2.30; 2.3; 2.30 hours; 2.30 min; 2 h 3; 2.3 h (this is because this indicates 0.3 of an hour (i.e. 18 minutes) rather than 30 minutes)
Any correct conversion with appropriate units, e.g. 2.5 hours; 150 mins unless the question specifically asks for time given in hours and minutes	02:30 (as this is a 24-hour clock time, not a time interval) 2.5; 150

Time

The table below gives guidance for answers involving time.

The table shows acceptable and unacceptable versions of the answer 07:30

	Accept	Do not accept
If the answer is required in 24-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 07:30 with any separator in place of the colon, e.g. 07 30; 07,30; 07-30; 0730	7:30 7:30 am 7 h 30 m 7:3 730 7.30 pm 073 07.3
If the answer is required in 12-hour format	Any unambiguous indication of correct answer in numbers, words or a combination of the two, e.g. 7:30 am with any separator in place of the colon, e.g. 7 30 am; 7.30 am; 7-30 am 7.30 in the morning Half past seven (o'clock) in the morning Accept am or a.m.	Absence of am or pm 1930 am 7 h 30 m 7:3 730 7.30 pm

Algebra

The table shows acceptable and unacceptable versions of the answer $3x - 2$

Accept	Do not accept
$x^3 - 2$; $3 \times x - 2$	$3x + -2$ if it is supposed to be in simplest form
Case change in letters	
Changes in letters as long as there is no ambiguity	

Accept extra brackets when factorising, e.g. $5(x + (3 + y))$

Teachers must mark the final answer given. If a correct answer is seen in working but final answer is given incorrectly then the final answer must be marked. If no answer is given on the answer line then the final line of the working can be taken to be the final answer.

Inequalities

The table shows acceptable and unacceptable versions of various answers.

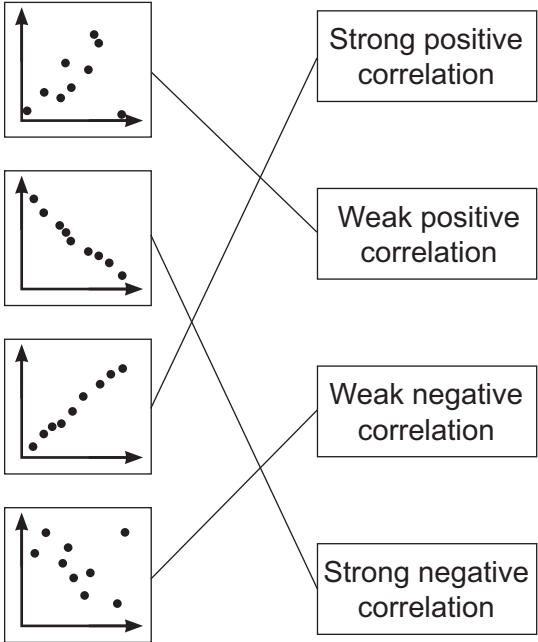
For the following	Accept	Do not accept
For $6 \leq x < 8$	$[6, 8)$	$< x <$
For $x \leq -2$	$(-\infty, -2]$	$x < -2$
For $x > 3$	$(3, \infty)$ $3 < x$	Just '3' written on the answer line, even if $x > 3$ appears in the working

Plotting points

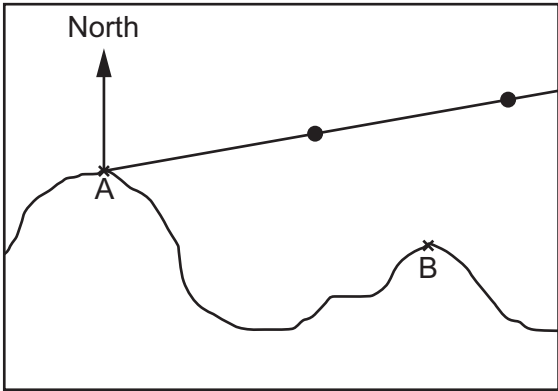
The table shows acceptable and unacceptable ways to plot points.

Accept	Do not accept
Crosses or dots plotted within $\pm \frac{1}{2}$ square of the correct answer	A horizontal line and vertical line from the axes meeting at the required point
The graph line passing through a point implies the point even though there is no cross	

Question	Answer	Marks	Part Marks	Guidance
1	7.6(3...) (cm)	2	Award 1 mark for 6.88(2) (cm) or 6.9 (cm) after 1 week or for any fully correct working, e.g. $6.2 \times 1.11 \times 1.11$ (may be seen in stages).	Accept better for 2 marks, e.g. 7.64
2	$(4n)^2 + 6$ or $16n^2 + 6$	1		
3	$\sqrt{8100}$ $\left(\frac{22}{7}\right)$ (7) $\sqrt{7}$ $\sqrt[3]{8}$	1		Accept any clear indication. All four answers correct for the mark.
4	$y = 0.5x + 3$ or $y = \frac{1}{2}x + 3$ or equivalent	2	Award 1 mark for $y = mx + 3$, $m \neq 0.5$ or equivalent or $y = 0.5x + c$, $c \neq 3$ or equivalent or $0.5x + 3$ or equivalent.	
5	Mia ticked and Gives correct explanation, e.g. (The other answer) is to 2 significant figures. or (The other answer) is to 3 decimal places. or 6 is the 3rd significant figure. or 7 is the 2nd significant figure. or 2 is the 1st significant figure. or The 0 after the decimal point is not a significant figure.	1		Both Mia ticked and any correct explanation needed for the mark.

Question	Answer	Marks	Part Marks	Guidance
6	 Strong positive correlation Weak positive correlation Weak negative correlation Strong negative correlation	1		All four lines correct for the mark.
7	0.57	1		
8	8 (hours)	1		

Question	Answer	Marks	Part Marks	Guidance																																			
9(a)	(Midpoint, x) 9 and ($f \times x$) 45 and (Total) 134	1		All three answers correct for the mark.																																			
9(b)	6.7 (kg)	1		Follow through <i>their</i> total $f \times x$ correctly divided by 20																																			
9(c)	Exact values are not known. or The data is grouped. or Midpoints have been used.	1		Accept any correct explanation.																																			
10	Correct stem-and-leaf diagram drawn. <table><tr><td>0</td><td>8</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>6</td><td>7</td><td>9</td><td></td><td></td><td></td></tr><tr><td>2</td><td>0</td><td>3</td><td>7</td><td>7</td><td>8</td><td>9</td></tr><tr><td>3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>1</td><td>4</td><td></td><td></td><td></td><td></td></tr></table> Key: 1 6 represents 16 people	0	8						1	6	7	9				2	0	3	7	7	8	9	3							4	1	4					3	Award 3 marks for correct stem-and-leaf diagram. or Award 2 marks for one error. or Award 1 mark for two errors.	One error, e.g. Incorrectly ordered leaves in one row. or Not including all five stems. or Not including all 12 leaves. or Incorrect key.
0	8																																						
1	6	7	9																																				
2	0	3	7	7	8	9																																	
3																																							
4	1	4																																					
11	7 8 9 10	1		Accept any clear indication.																																			

Question	Answer	Marks	Part Marks	Guidance
12(a)	(They all have) gradient of 3	1		
12(b)	(They all have) intercept on y -axis of -2	1		Accept any correct description of intercept on y -axis.
12(c)	$y = -2x + 3$ placed in $\mathcal{E}$ outside of A or B	1		
13	Two correct positions for ship S marked on the map (bearing from A of 080° and distance from B of 3.6 cm). 	3	Award 2 marks for one correct position for ship S marked on the map. or Award 1 mark for a ship S marked on a bearing of 080° or for a ship S marked at a distance of 3.6 cm from B or for a correct calculation of 3.6 cm.	Tolerance ± 2 mm and 2° .
14	$x^2 - 3x - 28$	2	Award 1 mark for three correct terms from $x^2 - 7x + 4x - 28$	Note $-3x$ counts as two terms.
15	21	1		

Question	Answer	Marks	Part Marks	Guidance
16	$(x =) \frac{5}{9}(y - 4)$	2	Award 1 mark for $y - 4 = \frac{9}{5}x$ or $5y = 9x + 20$ or $\frac{y}{9} = \frac{1}{5}x + \frac{4}{9}$	Accept equivalent answers for 2 marks, e.g. $(x =) \frac{5y - 20}{9}$
17(a)	3.5 and 2 and -2.5	2	Award 1 mark for one or two correct values.	
17(b)	Straight line joining $(-1, 3.5)$ to $(3, -2.5)$.	1		
18	980 (cm ²)	3	Award 1 mark for $\pi \times 12 \times 20$ or better. and Award 1 mark for area of circle $\pi \times 6^2$ or better.	Accept values in range of 979 to 981 for 3 marks. Accept values for π in the range of 3.14 to $\frac{22}{7}$ Or better, e.g. 753.9... Or better, e.g. 113.0...

Question	Answer	Marks	Part Marks	Guidance
19(a)	0.6 for chapattis and 0.5 for all drink branches	2	Award 1 mark for 0.6 for chapattis or 0.5 for all drink branches.	Accept equivalent fractions and percentages.
19(b)	0.3	2	Award 1 mark for <i>their</i> P(chapattis) $\times$ <i>their</i> P(tea).	Accept equivalent fractions and percentages. <i>Their</i> P(chapattis) and <i>their</i> P(tea) must be between 0 and 1
20	48 (cm ²)	2	Award 1 mark for $54 \div 18$, implied by 3 or $16 \times \text{their } 3$	
21	$43\,450 \leq \text{Eva's number} < 43\,500$	2	Award 1 mark for one correct limit or for 42 500 and 43 500 (limits for 43 000) and 43 450 and 43 550 (limits for 43 500).	
22	59.2 (cm ²)	4	Award 3 marks for complete method, i.e. $\frac{3}{4} \times \pi \times 4.2^2 + 4.2^2$ or Award 2 marks for $\frac{3}{4} \times \pi \times 4.2^2$ or Award 1 mark for $\pi \times 4.2^2$ and If 3 marks not scored, award 1 mark for 4.2×4.2 or for rounding <i>their</i> more accurate area correctly to one decimal place.	3 marks implied by answers which round to 59.2 (cm ²).

Question	Answer	Marks	Part Marks	Guidance
23	76.8 (cm ³)	4	Award 2 marks for $\sqrt{13.3^2 - 12.1^2}$ or Award 1 mark for $x^2 + 12.1^2 = 13.3^2$ or better In addition, award 1 mark for <i>their AB</i> from attempt at Pythagoras $\times 12.1 \times 0.5 \times 2.3$	Accept 77 with correct working for 4 marks. 2 marks implied by 5.52... Or better for 1 mark, e.g. $13.3^2 - 12.1^2$ or 30.48 An attempt at Pythagoras must involve 13.3^2 and 12.1^2, but these may be added rather than subtracted.

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